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Sustainable change.

Sample survey Hydrogen market design



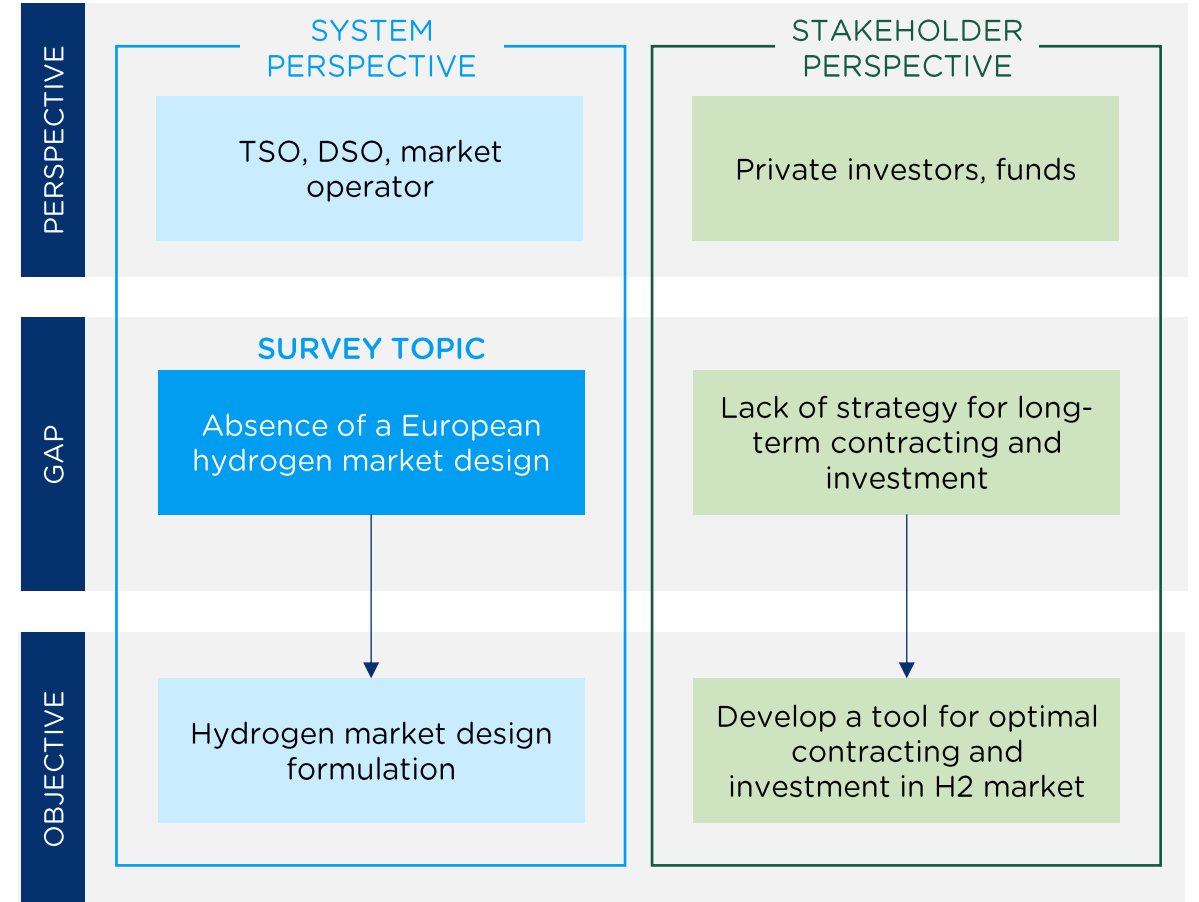
Agenda for today

1. Project introduction
2. Interview questionnaire
3. Further remarks

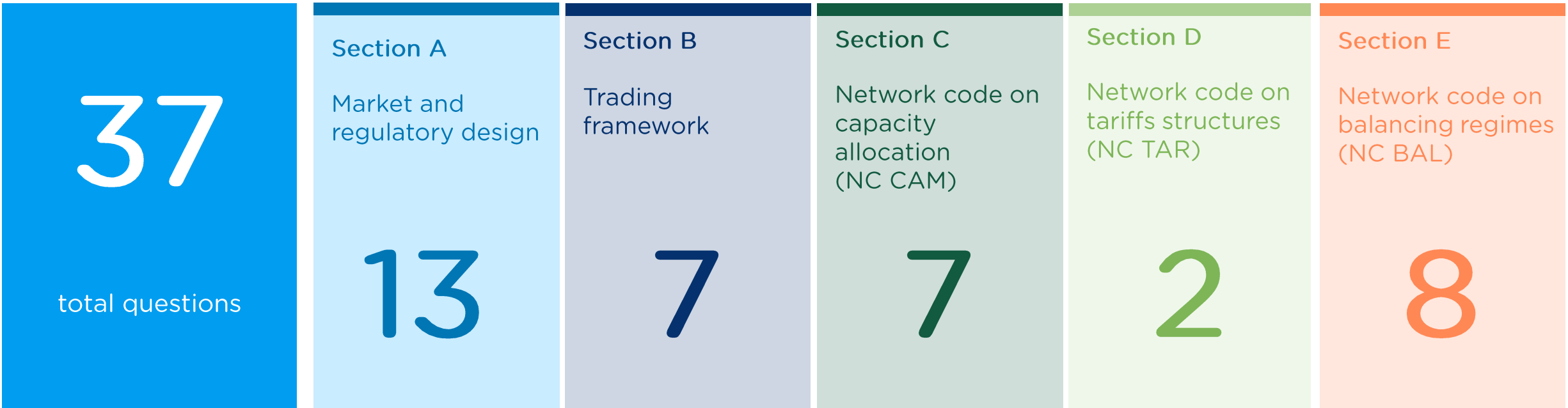
01. Project introduction

This PhD project will investigate the commoditization of green hydrogen in Europe

Project title	Commoditizing green hydrogen in Europe: From efficient market design to optimal contracting and investment
PhD Candidate	Marco Saretta
Partner company	Rambøll Denmark A/S
Partner university	Technical University of Denmark (DTU)



Survey overview



02. Questionnaire

1. Market and regulatory design
2. Trading framework
3. Network Code on Capacity Allocation Mechanisms (NC CAM)
4. Network Code on Tariffs (NC TAR)
5. Network Code on Balancing (NC BAL)

1st Section – Market and regulatory design - 1/2

1. Who is the regulating authority in your country?
2. Do you have a regulation scheme in place? If so, when is it expected to be implemented?
3. Will the entry/exit point scheme be maintained in the hydrogen market?
4. Will the gas day and gas year structure be kept for hydrogen?
5. How will the shipper roles evolve, especially at early market stages with lower liquidity?
6. In your country, how would you expect the hydrogen generation to be? Green, blue, or gray?
7. How will Guarantees of Origin play a role with RFNBO?

1st Section – Market and regulatory design - 2/2

1. What technical and regulatory challenges of the new hydrogen economy are specific to your TSO's country?
2. What is the hydrogen strategy for the new hydrogen economy in your TSO's country?
3. What are the expectations regarding interaction with TSOs within the same country and neighboring countries?
4. Which role will DSOs play at the early market stage, given the large industrial nature of H₂ off-takers?
5. What in your opinion has the most urgent requirement to be developed?
6. If you had time to do research, what would be some challenges that you wish you could solve for future hydrogen markets?

2nd Section – Trading Framework

1. Should we expect an exchange for H₂ (like ICE)?
2. Should we expect a virtual trading hub for H₂ (like TTF)?
3. Should we expect H₂ to be traded at exchanges, via OTC trades, or both, as in natural gas?
4. How will the price formation mechanism function?
5. How correlated will hydrogen prices and electricity prices be?
6. Which role will derivative contracts (futures, forwards, PPAs) play in electricity provision and hydrogen intake?
7. What will be the impact of CO₂ quotas?

3rd Section – Network code on capacity allocation (NC CAM)

1. Will the scheme of nominations, re-nomination, and allocations be kept for hydrogen networks?
2. How will the gate closure time for nomination submissions in the gas day be affected?
3. Will the capacity products differ from natural gas markets (e.g., yearly, quarterly, monthly, daily, intraday)?
4. What is the reasoning behind the capacity products you offer?
5. How much capacity is expected to be available in the early stages?
6. How and when are you scheduled to go live with the capacity?
7. Will the TSO assign capacity with mechanisms different from ascending clock auctions?

4th Section – Network code on tariffs structures (NC TAR)

1. How will the capacity base price and premium be decided? (System tariffs)
2. Will the current congestion management system need adjustments for hydrogen?

5th Section – Network code on balancing regimes (NC BAL)

1. Do you have a network code on balancing which will govern the framework for the balancing model?
2. How will the hydrogen target balancing model differ from the natural gas one?
3. What is the expected role of linepack in hydrogen markets? Can you quantify it?
4. Will the determination methodology for green, yellow, and red zones be adjusted in the future H₂ market?
5. How will continuous balancing work for hydrogen markets?
6. How can the duality between storage needs and the lack of availability at early market stages be addressed?
7. How will shippers be incentivized to maintain an optimal balance in the network?
8. How will shipper deal with variable supply and constant offtake?

03. Further remarks

Final remarks

We would kindly ask:

- What important questions did we forget in the questionnaire?
- Contact of relevant TSO to interview with?



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